

# As-Needed Treatment of Postoperative Hypertension for Inpatient Surgical Patients

## A Review

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**IMPORTANCE** Asymptomatic postoperative hypertension is a common clinical condition in admitted surgical patients. As-needed treatment with medication to acutely lower measured systolic blood pressure can have adverse outcomes.

**OBSERVATIONS** Existing literature and current practice guidelines indicate that treating asymptomatic hypertension can lead to adverse clinical outcomes. Because of systematic triggers, including nursing notifications, unfamiliarity with perioperative medical studies, and as-needed medication orders, it is often treated in the postoperative period.

**CONCLUSIONS AND RELEVANCE** Postoperative hypertension should only be treated when it is symptomatic. A conscientious effort should be made to minimize systematic triggers that lead to unnecessary treatment.

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**H**ypertension is a common long-term medical condition in the United States, affecting nearly half of the adult population.<sup>1</sup> In the United States, hypertension is defined as a systolic blood pressure (SBP) greater than 130 mm Hg and diastolic blood pressure (DBP) greater than 80 mm Hg. Globally, the majority of nations define hypertension using a higher threshold of 140/90 mm Hg, as outlined in the International Society of Hypertension and the European Society of Cardiology/European Society of Hypertension.<sup>2-4</sup> Management guidelines for patients in the outpatient setting are well studied and supported by robust data demonstrating reduced complications associated with longstanding hypertensive disease. However, treatment of high blood pressure in the inpatient setting is less systematically studied. This is particularly true for postoperative patients, among whom hypertensive blood pressure measurements are common.

The cause of postoperative hypertension is often multifactorial, attributable to pain, stress, anesthetic agents, temperature dysregulation, hypoxia, volume shifts, and/or the cessation of daily oral antihypertensive medications. As a result, hypertension is often identified and treated by inpatient surgical and nonsurgical clinicians. However, the acute lowering of asymptomatic high blood pressure in patients who often have long-term hypertension is not robustly supported by the literature and may, in fact, lead to more complications.

This review will summarize best-practice guidelines for the management of postoperative hypertension, related studies, and potential consequences of overtreatment. Although no robust clinical trials have addressed this question, observational studies, meta-analyses, and national association guidelines are used.

### Prevalence

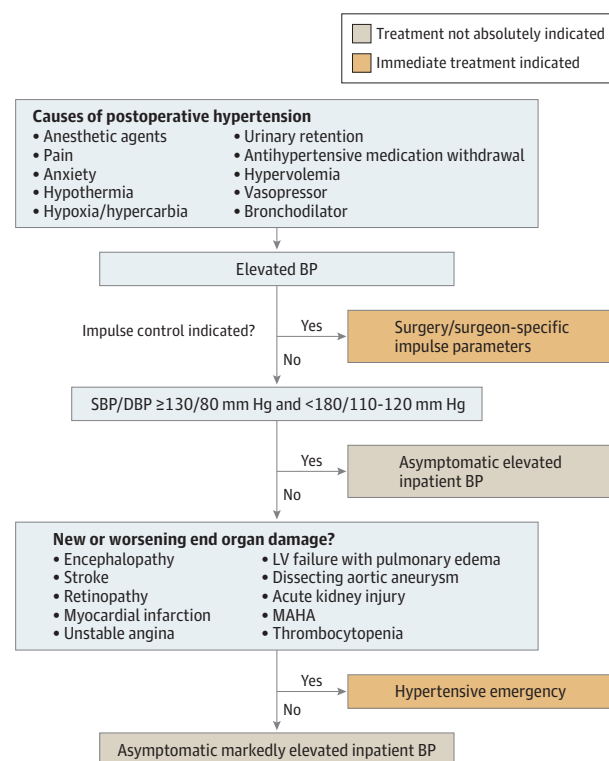
Up to 72% of inpatients will have hypertension during acute care admissions, and 1 in 4 patients undergoing major noncardiac

surgery will experience hypertension in the perioperative period.<sup>5,6</sup> A prospective observational study continuously monitored 302 patients after abdominal surgery and found that 40% of them spent at least 30 continuous minutes with mean arterial pressures above 110 mm Hg, which corresponds to a cuff pressure typically greater than 170/80 mm Hg.<sup>7</sup> The majority (73%) of these episodes of hypertension were missed with routine nursing vital sign monitoring (as opposed to the continuous monitoring in the study), indicating that many episodes of postoperative hypertension likely go unidentified. In fact, some studies report identified rates of acute postoperative hypertension as low as 10%, suggesting that accurate identification of hemodynamic shifts is poor in the inpatient setting.<sup>8</sup> While the exact incidence of postoperative hypertension is unknown, and measurement quality and intervals vary, hemodynamic lability is a common clinical presentation among surgical patients. Transient postoperative hypertension has not been demonstrated to influence clinically meaningful outcomes. While data are limited on this topic, most complications (eg, cardiac events) arising from perioperative hypertension have been associated with intraoperative hypertension or sustained hypertension.<sup>9</sup>

### Pathophysiology

The cause of postoperative hypertension is multifactorial (**Figure**).<sup>8</sup> Sympathetic activation due to surgical pain may lead to increased heart rate and blood pressure. Similarly, psychological stress may cause autonomic responses that lead to increased blood pressure. "White coat hypertension" has been a well-documented physiologic process in which patients experience higher than baseline blood pressure readings as a result of anxiety associated with health care visits.<sup>10</sup>

**Figure. Flowchart of Decision-Making for Elevated Blood Pressure (BP) Findings Among Inpatients**



Definitions as proposed by the American Heart Association Scientific Statement in 2024.<sup>6</sup> DBP indicates diastolic blood pressure; LV, left ventricle; MAHA, microangiopathic hemolytic anemia; SBP, systolic blood pressure.

Beyond autonomic responses, volume shifts associated with surgery and perioperative resuscitation can lead to blood pressure abnormalities. While hypotension is often the concern for a postoperative patient, significant volume replacement may have the opposite effect and result in hypertension. Other postoperative homeostatic derangements, such as temperature extremes, urinary retention, immobility, and operative site (eg, perianal, gynecologic), can lead to increased blood pressure measurements as well.<sup>11</sup>

Furthermore, more than 25% of the adult US population reports taking antihypertensive medications.<sup>12</sup> In the perioperative setting, these are often stopped to prevent postoperative hypotension, which may occur as a result of blood and insensible losses, anesthetic agents, and prolonged time fasting. Stopping angiotensin-converting enzyme inhibitors (ACEi) and angiotensin receptor blockers (ARB) is also done frequently in the perioperative period to prevent postinduction hypotension and protect kidney function.<sup>11</sup> However, cessation of a patient's long-term antihypertensive medications may lead to rebound hypertension before and after surgery.

## Clinical Presentation

Postoperative hypertension can be asymptomatic or symptomatic, and that difference is critical in clinical decision-making. The American Heart Association (AHA) has proposed the follow-

ing terminology (Figure): hypertensive emergency (SBP/DBP  $>180/110-120$  mm Hg with evidence of new or worsening target-organ damage), asymptomatic markedly elevated inpatient blood pressure (SBP/DBP  $>180/110-120$  mm Hg without evidence of new or worsening target-organ damage), and asymptomatic elevated inpatient blood pressure (SBP/DBP  $\geq 130/80$  mm Hg without evidence of new or worsening target-organ damage).<sup>6</sup>

The new or worsening target-organ damage that indicates hypertensive emergency may include hypertensive encephalopathy or evidence of stroke, acute myocardial infarction or ischemia, unstable angina, aortic aneurysm dissection, acute kidney injury, or thrombocytopenia. Therefore, any evidence of new or worsening target organ damage, or even symptomatic hypertension, should be promptly evaluated by a bedside physician and treated if clinically indicated. Markedly elevated blood pressure without these or other clinical findings indicative of organ damage does not reflect a hypertensive emergency and is far more common in the inpatient setting than episodes of true hypertensive emergency.

## Assessment and Diagnosis

The assessment and diagnosis of postoperative hypertension consists of first evaluating for new end-organ damage to rule out a hypertensive emergency (Figure). This should include standard vital signs and patient evaluation for acute neurologic or cardiac changes on physical examination and electrocardiogram or telemetry monitoring. If a hypertensive emergency is identified, it should be promptly treated to avoid end-organ damage.

Criterion standards of blood pressure measurement have been outlined by the AHA for the outpatient setting, including optimal cuff selection and patient positioning, but no corollary guidance exists for inpatient measurement.<sup>13</sup> In the inpatient setting, blood pressure is often measured by automatic oscillometric devices. A single measurement with an automatic blood pressure cuff can be inaccurate, with error ranges for SBP from  $-14$  to  $33$  mm Hg and for DBP from  $-6$  to  $19$  mm Hg.<sup>14</sup> Multiple factors, such as cuff size, measurement technique, nonvalidated or poorly calibrated devices, and environmental factors, can affect readings.<sup>14-16</sup> Numerous patient factors have also been identified to alter accurate blood pressure measurement, including temperature, bladder distention, recent food or beverage intake, white coat effect, body positioning, clothing, and cuff placement.<sup>17</sup> The accuracy of a single reading is most often insufficient for clinical decision-making, and guidelines recommend averaging multiple measurements to minimize error.<sup>15</sup> In addition, machines can underestimate true blood pressure, particularly when SBP is higher than  $180$  mm Hg.<sup>18</sup>

Protocolized order sets clearly influence and guide perioperative management, particularly when based solely on numerical values and lacking clinical assessment. For example, pain management order sets based on numeric-only ratings without clinician assessment found that the incidence of opioid oversedation adverse drug reactions more than doubled (from  $11.0$  to  $24.5$  per  $100\,000$  inpatient days), and that  $94\%$  of these adverse events were preceded by a documented decrease in level of consciousness, underscoring the importance of clinical assessment beyond pain scores alone.<sup>19</sup> The authors concluded that relying exclusively on pain scores

for opioid administration compromises patient safety and increases the risk of overmedication.

Furthermore, in the modern inpatient setting, the ease with which nursing staff communicates with clinicians, defined as physicians, nurse practitioners, or physician assistants, through electronic medical health programs or electronic pages can lead to immediate communication of elevated blood pressures and the potential overtreatment of abnormalities. An average of 34 messages are sent per day for every inpatient, with 20% of messages from nurses to inpatient clinicians.<sup>20,21</sup> A borderline elevated reading on an automatic cuff may prompt a message to the covering clinician requesting an as-needed medication order or a 1-time intravenous agent without contextualization of patient factors, such as timing of reading, concurrent pain or agitation, cuff size, fit or placement, and absence or presence of associated signs or symptoms of hypertensive emergency.

## Treatment

The primary postoperative hypertension management strategy should be prevention. The only exceptions are surgical procedures that warrant tight postoperative hemodynamic impulse control with parenteral agents, for example, vascular and cardiac interventions. In these clinical scenarios, tight blood pressure control is indicated to prevent severe postoperative complications and bleeding. These include neurosurgical interventions and carotid endarterectomy to prevent intracranial hemorrhage, aortic procedures to limit aortic wall stress, and cardiac surgical procedures to reduce postoperative arrhythmias and reduce peripheral vascular resistance. American College of Cardiology (ACC) guidelines recommend restarting preoperative antihypertensive medications as soon as clinically feasible.<sup>22,23</sup> Propensity-matched Veterans Affairs cohort studies, a population enriched for long-term cardiovascular disease, found that delaying resumption of preoperative ACEi/ARBs was associated with increased 30-day mortality risk.<sup>24,25</sup> Importantly,  $\alpha$ -1 antagonists,  $\beta$ -blockers, and clonidine, if stopped, can cause rebound hypertension, and special attention needs to be given to these patients when writing postoperative orders.

Many medical personnel are not familiar with the impact of patient factors and device limitations on inpatient blood pressure measurements as outlined above. Therefore, asymptomatic elevated blood pressures measured by automatic devices are often treated, both directly and indirectly, by on-call staff and house officers. It is not uncommon for postoperative orders to have standing as-needed intravenous antihypertensive medication orders for SBP at a certain level. In addition, nursing pages or texts for asymptomatic blood pressures are standard practice. One retrospective cohort study of Veterans Affairs patients found that 21% received as-needed blood pressure medications, and use of these medications was associated with increased risk of acute kidney injury.<sup>26</sup> Another study examined patients from acute care surgery and orthopedic surgery services at a single quaternary referral center and found that more than 10% of patients (114/1071) received a total of 552 as-needed antihypertensive administrations, most often for an arbitrary SBP threshold above 160 mm Hg.<sup>27</sup> Labetalol was most frequently prescribed in this study, followed by hydralazine. Hydralazine is specifically recommended not to be used in the 2025 ACC/AHA

### Box. Reasons for Overtreatment of Asymptomatic Elevated Blood Pressure

- Incorrect measurement, including confounding patient factors
- Protocolized order sets
- Standing as-needed orders
- Virtual communication of measurement without concurrent patient assessment
- Unfamiliarity with published blood pressure management guidelines

Joint Committee Clinical Practice Guidelines.<sup>23</sup> The drug lowers blood pressure within 10 to 30 minutes, with effects lasting 2 to 4 hours. The guideline specifically states, "Hydralazine is an undesirable first-line agent for acute treatment in most patients due to unpredictability of response and prolonged duration of action,"<sup>23</sup> yet this is one of the most commonly ordered medications, likely because it can be given by a single injection and does not require the medication to be given as an infusion. Among patients with orders for as-needed intravenous antihypertensive medications, those who were actually given the medications had longer hospitalizations.<sup>28</sup> When surveyed, house officers and hospitalists reported that asymptomatic elevated blood pressures, SBP between 180 and 190 mm Hg, would prompt them to delay discharge for a patient.<sup>29</sup>

Postoperative patients often have their postoperative orders and subsequent care directed by surgeons. This has both negative and positive implications. Residents of surgical specialties are less likely to be familiar with or follow published blood pressure management guidelines but are also less likely to change a patient's standing antihypertension regimen. For an SBP greater than or equal to 160 mm Hg, 57% of internal medicine residents reported that they would change a patient's regimen, whereas only 29% of surgical residents would do so.<sup>30</sup>

If a patient has a hypertensive emergency, recommended treatment regimens are based on the suspected or identified type of end-organ damage and include labetalol, nicardipine, nitroprusside, clevidipine, hydralazine, esmolol, metoprolol, and nitroglycerin.<sup>31</sup> It is recommended to decrease blood pressure by no more than 20% to 25% in the first hour if end-organ damage is suspected or confirmed.<sup>32</sup> Overtreatment of elevated blood pressure that then causes hypotension can occur approximately 10% of the time, with the greatest risk of hypotension occurring in the first 6 to 12 hours after use of an intravenous agent.<sup>33-35</sup>

## Adverse Outcomes

There are no randomized clinical trials supporting the treatment of asymptomatic hypertension in the inpatient setting. Yet, because of bedside automatic cuff readings, monitor alarms, nursing pages, and as-needed medication orders, medical intervention is usually undertaken (Box). A 2024 scientific statement from the AHA reviewed existing literature in this space. It recommended that permissive hypertension should be considered for patients in acute care settings (inpatient and emergency department) who have an SBP greater than 180 mm Hg without evidence of end-organ damage.<sup>6</sup>

Table. Consequences of Inpatient Hypertension Overtreatment

| Source                                 | Purpose                                                                                                     | Methods                                      | Select results                                                                                                                    |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Weder and Erickson, <sup>33</sup> 2009 | Frequency of IV antihypertensive medication use and increased LOS                                           | Retrospective single-institution study       | Increased LOS for those receiving IV antihypertensive medication (12.0 d vs 7.1 d for hydralazine, 11.8 d vs 7.9 d for labetalol) |
| Mudumbai et al, <sup>24</sup> 2014     | Evaluate 30-d mortality risk with postoperative nonresumption of ACEi                                       | Retrospective cohort study, Veterans Affairs | 30-d Mortality rate increased with nonresumption (HR, 3.44)                                                                       |
| Lee et al, <sup>25</sup> 2015          | Evaluate failure to resume ARB impact on all cause 30-d mortality                                           | Retrospective cohort study, Veterans Affairs | 30-d Mortality 1.3% if ARB resumed by postoperative day 2, 3.2% if not (HR, 2.45)                                                 |
| Lizano-Díez et al, <sup>36</sup> 2022  | Consequences of acute hypertensive/hypotensive episodes                                                     | Systematic review                            | 1.51 Odds higher AKI for 10% decrease in MAP; 10 min of postoperative hypertension associated with greater 30-d mortality         |
| Anderson et al, <sup>37</sup> 2019     | Evaluate association between intensification of antihypertension regimen at discharge and clinical outcomes | Retrospective cohort                         | Intensification associated with higher risk of readmission and SAEs                                                               |
| Canales et al, <sup>31</sup> 2025      | Risk of AKI from as-needed BP administration                                                                | Retrospective cohort study, Veterans Affairs | Increased AKI risk (HR, 1.23)                                                                                                     |

Abbreviations: ACEi, angiotensin-converting enzyme inhibitor; AKI, acute kidney injury; ARB, angiotensin receptor blocker; BP, blood pressure; HR, hazard ratio; IV, intravenous; LOS, length of stay; MAP, mean arterial pressure; SAEs, serious adverse events.

Observational studies have demonstrated that among patients receiving new antihypertensive medications during an inpatient admission, hypotensive episodes were more common and resulted in higher rates of acute kidney injury, myocardial infarction, and stroke, compared with patients whose hypertension was untreated (Table).<sup>12,24,25,31,33,36,37</sup> Relatedly, there is a demonstrated 1.51 higher likelihood of acute kidney injury or additional kidney complications for every 10% decrease in mean arterial pressure.<sup>36</sup> A meta-analysis studying perioperative hemodynamic abnormalities reported that postoperative hypotension for more than 10 minutes is associated with greater odds of 30-day mortality.<sup>36</sup>

Therefore, overtreatment of asymptomatic hypertension is not without risks. Hemodynamic lability is associated with adverse outcomes, including increased 30-day operative mortality, greater risk of postoperative cardiac death, and perioperative complications.<sup>38-40</sup> Among older adults with hypertension, even intensifying outpatient regimens after hospitalization has been associated with increased risk of readmission and serious adverse events.<sup>37,41</sup>

### Limitations

Given the heterogeneity in inpatient and postsurgical care, most studies included are observational. While these data are largely observational, a clear signal is seen throughout the body of literature to indicate the detriments of postoperative hypertension overtreatment. Furthermore, given the limited published data on the topic, a narrative form search strategy was used, aiming to include all relevant publications discussing the inpatient management of postoperative hypertension. We took particular care to only include re-

search done since the advent of automatic measurement, which first became standard of care in the 1990s.<sup>41,42</sup> In addition to the increased use of automatic measurement devices, we believe that the incremental lowering of hypertension definition thresholds, now down to SBP less than 130 mm Hg in the 2017 ACC/AHA guidelines, has led to an increased emphasis on acute blood pressure lowering in the inpatient setting.<sup>43</sup> Therefore, observational studies and societal guidelines from the last 10 years were considered pivotal in this review.

### Conclusions

While most data reviewed here are observational, there appear to be significant consequences and potential iatrogenic harm associated with treating asymptomatic hypertension in the inpatient setting, particularly among postoperative patients. The use of antihypertensive medications in the inpatient setting should be prudently evaluated and only for patients with evidence of hypertensive emergency as defined by the AHA in 2024. Increased education on the current classification of severe hypertension and treatment guidelines and early restarting of outpatient medications are critical for physicians and nursing staff. In addition, systematic changes should be made to limit the notification and overtreatment of asymptomatic hypertension in postoperative patients. Future prospective studies are needed to systematically evaluate the use of as-needed antihypertensive medications in postoperative patients and potential downstream consequences.

### ARTICLE INFORMATION

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